ROUND 1 REVIEW

AI Frontier Challenge 2026 Round 1 Submission

Team Name

StockSentinel AI

Team Leader

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Track

Retail

Date

15/10/2026

Use Case

Problem Statement

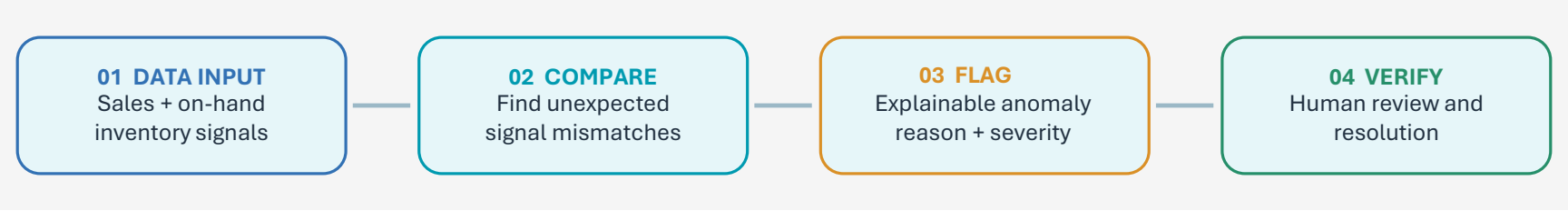
- Sales and recorded inventory can disagree, creating a phantom stock-out.
- A product may show positive stock while sales remain at zero for several days.
- This can lead to missed sales, manual checks, and poor replenishment decisions.

- Opportunity: retailers managing many SKUs across multiple stores.
- Users: store managers, retail operations teams, and inventory planners.
- Complements existing POS and inventory systems with a focused anomaly-review workflow.

Proposed Solution

Solution Overview

- StockSentinel AI flags suspicious SKU/store combinations.
- It compares sales history with recorded on-hand inventory.
- Explainable alerts, severity ranking, evidence, and a human verification queue help staff investigate.
- The system prioritizes review instead of silently changing stock records.



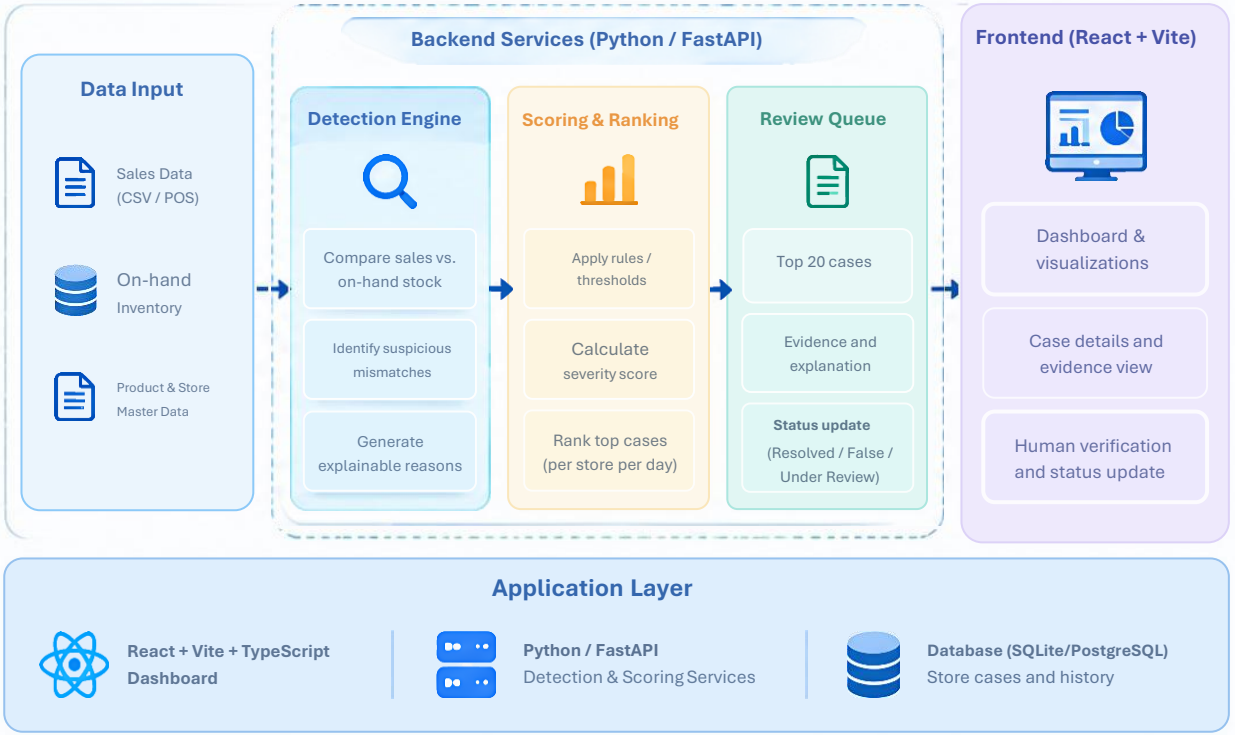
Output: a ranked case queue with evidence and a recommended next check.

Design Steps & Architecture

Technical Implementation

1. **Ingest Data**
   Load sales and on-hand inventory signals (CSV / system exports).
2. **Compare Signals**
   Validate and compare expected demand with observed sales.
3. **Detect Mismatches**
   Flag suspicious SKU-store cases with explainable reasons.
4. **Prioritize Cases**
   Rank by severity and likely business impact.
5. **Human Verification**
   Send ranked cases to a review queue for confirmation and resolution.

Architecture Diagram



Prototype uses demo/sample data. Secure API, persistent database, and POS/ERP connectors are future integrations.

Challenges & Mitigation

Challenge 1

Problem: Records may be missing, delayed, or inconsistent.
Mitigation: Validate required fields and dates; surface data-quality issues.

Challenge 2

Problem: Slow-moving products can naturally have zero-sales days.
Mitigation: Use historical demand context to reduce false positives.

Challenge 3

Problem: Promotions, holidays, and calendar gaps can distort demand signals.
Mitigation: Use available context and test edge cases; do not claim unsupported handling.

Unique Selling Proposition (USP)

Explainable alerts

Each alert explains which signals disagree and why the case was flagged.

Prioritized review

Ranks up to 20 suspicious cases per store per evaluation date, helping staff focus on the highest-priority investigations.

Human-in-the-loop

Staff inspect the evidence and record an outcome; the system does not silently alter inventory.

Competitive Advantage Summary

- • Faster triage, evidence attached to cases, and a clear next action.
- • Designed to complement existing systems rather than replace them.

Demo & POC Video

Live Demonstration

StockSentinel AI Demo Video

